WEB DEVELOPMENT

Lecture 1 — English Study Notes

Servers • Client–Server Architecture • Frontend & Backend • HTTP

Prepared from the lecture notes pasted in chat. The lecturer's terminology and framing are preserved as closely as possible, with organization added for revision.

Clickable Index / Contents

Click any topic below to jump to its section. The PDF viewer's bookmarks panel also provides quick navigation.

1. What a Server Is →
2. Why Servers Are Necessary →
3. Client–Server Architecture →
4. Frontend vs. Backend →
5. Frontend Technologies →
6. Backend Technologies →
7. HTTP Requests: URL and Method →
8. HTTP Responses: Status Codes →
9. Quick Revision Table →
10. Practice MCQs with Answers →
11. Final Revision Checklist →

Source note: the supplied material was a topic-wise outline, not a full verbatim transcript. Examples and MCQs clarify only the concepts included in that outline.

1. What a Server Is

A server is a special computer connected to the internet that hosts applications so people can access them.

Its basic architecture is similar to a local computer, but its capacity is much higher:

- Greater processing power.
- More RAM.
- Much larger storage capacity.

A server can hold both the application code and the database. It executes business logic such as login, logout, orders, and comments.

A local computer can technically act as a server, but it would need 24/7 uptime, reliable internet and power, protection from outages, and sufficient hardware.

Simple mental model

The client sends a request, the server processes it, and then the server returns a response.

2. Why Servers Are Necessary

The lecture gives two core reasons for hosting an application on a server rather than on a local machine:

Reason	Explanation
Accessibility	Keeping a local machine practically accessible to the public internet is difficult. A server makes the application accessible to users over the internet.
Scale	Large applications need to store and process huge amounts of data. The lecture gives Amazon and Facebook as examples, mentioning hundreds of thousands of terabytes of data.

Remember: servers help make applications accessible and capable of supporting large-scale usage.

3. Client–Server Architecture

In client–server architecture, the client sends a request. The server accepts and processes it, then returns a response.

Component	Role
End user	Uses the application through a browser or app.
Client (browser)	Sends requests to the server and displays the response on screen.
Server	Processes the request, executes business logic, and queries the database when needed.
Database	Stores the application's data.

Request and response flow

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End user → Browser / Client → Internet → Server → Database (if needed)
End user ← Browser / Client ← Internet ← Server ← Database result
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Important distinction: the end user and the client are not the same thing. The end user is the person; the browser is the client. The person interacts with the browser, and the browser interacts with the server.

An internet connection is required for client–server communication. This architecture is universal and is not specific to any particular technology.

4. Frontend vs. Backend

Frontend	Backend
Everything the user sees and interacts with: UI, navigation, buttons, and forms.	Processing, application logic, and data handling.
Runs or is displayed on the client side.	Consists of server-side application code and the database.

The backend has two parts:

- Application logic — server-side code that executes rules and processing.
- Database — where the application's data is stored.

A complete end-to-end application requires three pieces: UI + application logic + database.

Frontend and backend are equally important. Which one is more difficult depends on the feature and project being built.

5. Frontend Technologies

According to the lecture, the three core frontend technologies are HTML, CSS, and JavaScript.

Technology	Purpose	Easy way to remember
HTML	Creates the structure or skeleton of a web page.	Structure
CSS	Styles the page with colors, fonts, spacing, and layout.	Appearance
JavaScript	Adds client-side logic and interactivity, such as button-click behavior and form validation.	Behavior

Without CSS, a page may look plain and mostly black and white. HTML provides structure, CSS handles presentation, and JavaScript adds interaction.

Java and JavaScript are different programming languages. Despite the similar names, they are not the same language.

Libraries and frameworks

Libraries and frameworks build on existing technologies. The lecture compares them to a premix packet: ready-made functionality saves time instead of building everything from scratch.

Based on	Examples mentioned in the lecture
JavaScript	React.js, Angular, Vue.js, jQuery
CSS	Bootstrap, Tailwind CSS

6. Backend Technologies

Server-side logic can be written in different programming languages. The lecture lists Java, Python, .NET, PHP, C, and C++ as examples.

JavaScript can also run on the server side through Node.js. Therefore, one language can be used for both frontend and backend. This course will use JavaScript for both.

Database choices

Database examples mentioned in the lecture include MySQL, MongoDB, PostgreSQL, Oracle, and Cassandra.

A programming language does not have to be paired with one particular database. The technology stack is selected according to project requirements; IT architects make these decisions. This course will cover MySQL and MongoDB for the database layer.

Layer	Examples from the lecture
Frontend	HTML, CSS, JavaScript
Backend language	JavaScript for this course; Java, Python, .NET, PHP, C, and C++ are also mentioned
Database	MySQL and MongoDB for this course; PostgreSQL, Oracle, and Cassandra are other examples

7. HTTP Requests: URL and Method

According to the lecture, every client request has two mandatory components:

- URL / IP address — identifies which server should receive the request.
- Request method — specifies the purpose of the request.

HTTP method	Purpose described in the lecture	Example
GET	Retrieve data from the server.	Viewing existing product details
POST	Submit or save new data.	Submitting signup or registration data
PUT	Update existing data.	Updating an existing record
DELETE	Remove data.	Deleting a record

Other methods such as PATCH and OPTIONS also exist; the lecture says they will be covered later.

The request method is always present even when it is not directly visible to the user. The lecture mentions a live demonstration in the Network tab of browser DevTools on Flipkart.

Revision shortcut: GET = retrieve, POST = submit/save, PUT = update, DELETE = remove. This is the lecture's simplified summary of the methods.

8. HTTP Responses: Status Codes

Every server response includes a status code. Response data is optional, but the status is always present. It tells the client what happened on the server side.

A status consists of a numeric code and a predefined message. The key ranges from the lecture are:

Range	Meaning	Examples from the lecture
200–299	Success	200 OK
300–399	Redirection	No specific code listed
400–499	Client-side failure	400 Bad Request, 401 Unauthorized, 402 Payment Required, 404 Not Found
500–599	Server-side failure	500 Internal Server Error

Backend developers are responsible for sending the correct status code for each scenario. Status codes are predefined and universal; applications cannot redefine their standard meanings.

Quick distinction: 4xx vs. 5xx

The 4xx range generally indicates a client-side issue, such as a problem with the request or authentication. The 5xx range indicates a server-side failure. The exact meaning depends on the specific status code.

9. Quick Revision Table

Question	Quick answer
What is a server?	An internet-connected, high-capacity computer that hosts applications.
Why are servers needed?	Accessibility and scale.
Example of a client?	A web browser.
End user vs. client?	The end user is a person; the client is the browser or software.
Three core frontend technologies?	HTML, CSS, JavaScript.
Role of HTML?	Structure.
Role of CSS?	Styling and presentation.
Role of JavaScript?	Logic and interactivity.
Parts of the backend?	Application logic and database.
Backend language for this course?	JavaScript for both frontend and backend.
Databases for this course?	MySQL and MongoDB.
Two request components?	URL/IP address and method.
GET / POST / PUT / DELETE?	Retrieve / submit-save / update / remove.
200–299?	Success.
300–399?	Redirection.
400–499?	Client-side failure.
500–599?	Server-side failure.

10. Practice MCQs with Answers

1. In client–server architecture, what is the browser?

- A. Database
- B. Client
- C. Server hardware only
- D. End user

Answer: B — The browser is the client; the end user interacts with it.

2. Which are the core frontend technologies?

- A. HTML, CSS, JavaScript
- B. Java, MySQL, CSS
- C. MongoDB, HTML, Python
- D. C, C++, Oracle

Answer: A — HTML, CSS, and JavaScript.

3. Which technology creates the structure of a web page?

- A. CSS
- B. JavaScript
- C. HTML
- D. MySQL

Answer: C — HTML creates the structure or skeleton.

4. Which technology adds interactions such as button-click behavior?

- A. JavaScript
- B. HTML only
- C. MySQL
- D. IP address

Answer: A — JavaScript adds client-side interactivity.

5. What are the two core reasons for hosting on a server?

- A. Fonts and colors
- B. Accessibility and scale
- C. HTML and CSS
- D. Login and logout

Answer: B — Accessibility and scale.

6. What are the two parts of the backend described in the lecture?

- A. HTML and CSS
- B. Browser and user
- C. Application logic and database
- D. URL and method

Answer: C — Application logic and database.

7. Which HTTP method retrieves data?

- A. GET
- B. POST
- C. PUT
- D. DELETE

Answer: A — GET.

8. Which method is listed for updating existing data?

- A. GET
- B. PUT
- C. OPTIONS
- D. DELETE

Answer: B — PUT.

9. Which range contains HTTP status code 404?

- A. 2xx
- B. 3xx
- C. 4xx
- D. 5xx

Answer: C — 404 Not Found is in the 400–499 range.

10. Which range includes 500 Internal Server Error?

- A. 2xx
- B. 3xx
- C. 4xx
- D. 5xx

Answer: D — It is a server-side failure.

11. Which two databases will this course cover?

- A. Oracle and Cassandra
- B. MySQL and MongoDB
- C. PostgreSQL and Oracle
- D. MongoDB only

Answer: B — MySQL and MongoDB.

12. Are Java and JavaScript the same language?

- A. Yes
- B. No

Answer: B — They are different programming languages.

11. Final Revision Checklist

- Revise the definition of a server, its capacity, and the need for 24/7 availability.
- Be able to explain accessibility and scale as reasons for using servers.
- Distinguish the roles of the client, end user, server, and database.
- Remember frontend vs. backend and the three pieces of a full end-to-end application.
- Review the roles of HTML, CSS, JavaScript, and the listed libraries/frameworks.
- Review backend language and database examples; course stack: JavaScript + MySQL/MongoDB.
- Revise the purposes of GET, POST, PUT, and DELETE.
- Remember the 2xx, 3xx, 4xx, and 5xx ranges and the examples listed.